

Question

9.34 g of elemental **A** was heated in concentrated HNO_3 , and after complete reaction, 12.48 g of solid **B** (an oxide of **A**) was obtained, with the release of reddish-brown gas (Reaction 1). **A** can also react with molten KOH in the presence of O_2 to yield solid **C** (Reaction 2), where the oxidation state of element **A** in **C** is lower than that in **B**. **B** is a commonly used oxidizing agent in organic chemistry. Treating the KOH solution of **B** with ammonia yields a light-yellow complex **D**.

Choose the correct option from the following.

- A.** The ratio of the sum of the stoichiometric coefficients of reactants to products in Reaction 1 is 7:11 (the coefficients are in their simplest integer ratio).
- B.** The oxygen content of **B** is 20.15%.
- C.** Reaction 2 produces two products in a 2:1 ratio.
- D.** The anion in **D** belongs to the T_d point group.
- E.** The potassium content in **C** is 23.52%.
- F.** All of the above options are incorrect.

Answer

Correct Answer: **E**

Detailed Explanation

The composition of **B** is AO_x , and the relative atomic mass of **A** is M_A . Then, $M_A/(M_A + 16x) = 9.34/12.48$. Simplifying, we obtain $M_A = 47.59x$. Enumerating the values in a table, when $x=4$, $M_A = 190.36$. Combining the fact that **B** is a commonly used oxidizing agent in organic chemistry, **A** is determined to be **Os**.

CHECKPOINT

1 PTS

A is **Os**

Since **B** is a commonly used oxidizing agent in organic chemistry, **B** is OsO_4 , with an oxygen content of 25.17%.

CHECKPOINT

1 PTS

B is OsO_4 , with an oxygen content of 25.17%, option B is incorrect

Reaction 1 simultaneously produces a reddish-brown gas, which is nitrogen dioxide. The reaction equation is: $\text{Os} + 8\text{HNO}_3 \rightarrow \text{OsO}_4 + 8\text{NO}_2 + 4\text{H}_2\text{O}$. The ratio of the sum of the coefficients of reactants to products in the simplest whole numbers is 9:13.

CHECKPOINT

1 PTS



CHECKPOINT

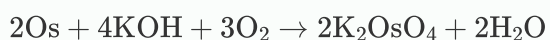
1 PTS

The ratio of the sum of the coefficients of reactants to products in the simplest whole numbers is 9:13, option A is incorrect

The oxidation state of Os in **C** is lower than that in **B**, which should be +6. The equation for reaction 2 is: $2\text{Os} + 4\text{KOH} + 3\text{O}_2 \rightarrow 2\text{K}_2\text{OsO}_4 + 2\text{H}_2\text{O}$. There are two products with a ratio of 1:1, so option C is incorrect.

CHECKPOINT

1 PTS



CHECKPOINT

1 PTS

Reaction 2 has two products with a ratio of 1:1, option C is incorrect

From reaction 2, the chemical formula of **C** is determined to be K_2OsO_4 , with a potassium content of 23.52%.

CHECKPOINT

1 PTS

The chemical formula of **C** is K_2OsO_4 , with a potassium content of 23.52%, option E is correct

Treating the KOH solution of OsO_4 with ammonia yields a yellow complex **D**. Based on elemental knowledge, the oxidation state of Os does not change in this process, resulting in KOsNO_3 . The anion has a tetrahedral geometry, but due to the unequal distribution of N and O, it belongs to the C_{3v} point group.

CHECKPOINT

1 PTS

The anion in **D** is a distorted tetrahedron, belonging to the C_{3v} point group, option D is incorrect